ISC Assignment-02

Jeetu Mehra

U24AI032

Question 1 : Implement a Caesar Cipher to encrypt and decrypt messages by shifting characters.

Examples:

```
Text  : ABCDEFGHIJKLMNOPQRSTUVWXYZ
Shift: 23
Cipher: XYZABCDEFGHIJKLMNOPQRSTUVW

Text  : ATTACKATONCE
Shift: 4
Cipher: EXXEGOEXSRGI
```

Code:

```
#include <iostream>
#include <string>
using namespace std;

int main() {

    // Original alphabet
    string alphabet = "ABCDEFGHIJKLMNOPQRSTUVWXYZ";

    int shift;
```

```
// Take shift as input
cout << "Enter shift: ";
cin >> shift;

// Keep shift within 0-25
shift = shift % 26;

// Create shifted alphabet
string shiftedAlphabet = "";

for (int i = 0; i < 26; i++) {
    shiftedAlphabet += alphabet[(i + shift) % 26];
}

// Display alphabets
cout << "\nOriginal Alphabet : " << alphabet << endl;
cout << "Shifted Alphabet  : " << shiftedAlphabet << endl;

// Take string to encrypt
string input;
cout << "\nEnter string to encrypt: ";
cin >> input;

// Encrypt the input
```

```
    string encrypted = "";

    for (char ch : input) {

        // Find character in original alphabet
        for (int i = 0; i < 26; i++) {

            if (ch == alphabet[i]) {
                // Replace with corresponding shifted character
                encrypted += shiftedAlphabet[i];
                break;
            }
        }
    }
    // Display encrypted string
    cout << "Encrypted String  : " << encrypted << endl;
    return 0;
}
```

Output:-

```
PS D:\PROGRAMMING\programms\ISC> cd "d:\PROGRAMMING\progr
if ($?) { g++ Q1.cpp -o Q1 } ; if ($?) { .\Q1 }
Enter shift: 3

Original Alphabet : ABCDEFGHIJKLMNOPQRSTUVWXYZ
Shifted Alphabet  : DEFGHIJKLMNOPQRSTUVWXYZABC

Enter string to encrypt: HITESH
Encrypted String  : KLWHVK
```